Independent review of the Topic C mathematical edition

Human Compression Project: independent review

1 Decision and boundary

This review accepts the specified four-application contract, subject to the exact artifact binding in the accompanying JSON record. The common paper and the four applications contain complete written arguments for that contract, with the retained mathematical inputs, corrected historical assertions and excluded auxiliary assertions stated explicitly. The two required finite programs are part of the proof of the Janko classification. The acceptance does not assert that every sentence of every source is recovered literally, that the whole volume is formalized in Lean, or that the rewritten arguments have historical priority. It makes no numerical compression claim and adds no source or question credit.

The historical boundary consists of the 35 previously accepted contract rows for L088, N013 and L108, and the five previously accepted F014 clauses. These include qualified and excluded rows; "40 rows" does not mean 40 newly proved theorems. The F014 higher-rank conjecture belongs to its separate admitted question packet and is not a result of this volume.

I read the complete common paper, all four applications, both required GAP programs and the three Lean modules. I read the complete available primary bodies for the four target sources. The L108 representation is the archived authored browser text of its seven printed pages, not an original PDF. Eleven critical pages of the three available source PDFs were also inspected visually. All 31 rendered pages of the six scientific/code documents were inspected. The complete primary and artifact inventory is recorded separately rather than redistributed with this review.

2 The common geometric arguments

The distinction between subgroup intersection and flag completion is preserved throughout. The coset-recognition proposition uses both the intersection property and typed coset Helly. A common coset representative completes a flag; the intersection property identifies its stabilizer. The panel and residual connectivity arguments then give a thin, residually connected geometry. The separate chamber-gallery argument proves that the full automorphism group acts freely on chambers. Consequently a chamber-transitive subgroup is the full group. None of these assertions is obtained by treating an arbitrary nonstring C-group as automatically defining a geometry.

The anchored Helly criterion is correct: right translation puts a previously intersecting family at the base parabolics, and its remaining coset is represented in a selected parabolic. The displayed product conditions are exactly the remaining pairwise intersections. In the finite counting form the represented cosets of the common subgroup already form a subset of the tested intersection; equality of their cardinalities therefore proves equality of those sets.

For endpoint conjugation, the proof lists every possible marked subgroup. Old–old and conjugated–conjugated intersections follow from the original intersection property. Mixed intersections use precisely the stated overlap identity. The only remaining intersections are between the groups containing both endpoint marks; the index-two dichotomy reduces these to intersection closure of the support family. Finite-

ness of the index set is now stated explicitly where the least-support conclusion uses it. No finite-group assumption is needed. The central-twist lemma preserves generation by recovering both commuting factors from their odd-order product, and it does not claim automatic preservation of the geometric conditions.

The graph-halving construction is a literal incidence construction. Higher objects retain a specified component in the bipartite double cover. At a fixed lifted vertex its residue is identified with the original residue. For a flag containing only higher objects the common shadow is either one component or both components of the lifted shadow graph; the precise parity condition is therefore necessary and sufficient for residual connectivity. The proof extends flags to chambers before deriving connectivity and does not identify a truncation with a residue without proof. Simplicity of the original graph makes an edge recoverable from its endpoints. This also supplies the faithful-action adapter and the endpoint-overlap condition. For lattice polytopes, the relevant shadows are nested, so the parity hypothesis is verified rather than omitted.

3 Binary modules and the inverse construction

For an arbitrary nonempty vertex set, the augmentation module uses finitely supported vectors. Edge differences along a path give the difference between its endpoints. The closed-walk relations make the reverse assignment independent of the chosen path, and the two compositions fix the edge generators and the augmentation basis. This proves the graph presentation even for infinite vertex sets; it does not require an unmentioned finite-dimensional argument.

The recognition theorem first constructs a homomorphism from the augmentation module using its basis. The covariance formula on that basis extends to the whole module. Multiplication then verifies directly that

$$\Psi(w, h) = T(w)j(h)$$

is a homomorphism from the semidirect product. Generation is an explicit hypothesis for surjectivity and for the second inverse identity. The conclusion is not supplied as a hypothesis: component maps and their stated values under the proposed forward map are checked separately.

I checked the cyclic displacement calculations with integer indices. The commutativity proof, both conjugation formulas, inversion by the reflection and the period reduction are valid. With $T_i = t_{-i}$, the formulas become exactly the augmentation-basis covariance for rotation and reflection. The relation $y = t_1 x^{-1}$ supplies generation. The inverse maps consequently prove equality with the presented group, not merely the existence of a finite quotient. The earlier finite-order upper-bound argument is no longer a required step. The displacement argument remains attributed to the source; standard augmentation and semidirect-product algebra is not described as a new invention.

As a separate finite diagnostic, I reconstructed the nonabelian example $S_3 = C_3 \rtimes C_2$ from explicit permutations and checked all 36 products and both inverse compositions. Further controls remove generation, covariance, or either component-inverse condition. These controls expose the need for the hypotheses; they are not substitutes for the general proof.

4 The four source applications

The first Janko group. The exact source presentation is identified by a completed coset enumeration with index 1463 over a subgroup of order at most 120, together with a quotient of order 175560. A valid finite permutation action alone would not establish the upper bound; the written proof and program correctly distinguish them. The transformed markings satisfy the required intersections and typed Helly conditions. The six multisets of maximal-parabolic orders are distinct, which separates geometries even when type

permutations are allowed. The two failed operations have the asserted nonthin panels. The figure-label transposition is kept separate from the definitions of the geometries.

The census proof retains the complete subgroup catalogue and exact algorithms, not the historical geometry census. Every rank-k residue of a target geometry lies inside some maximal proper subgroup, and its distinguished involutions occur in the local enumeration. Appending every involution outside the local generated subgroup is therefore exhaustive. The local group is not incorrectly required to equal its containing maximal subgroup. The rank-five stage retains proper subgroup targets too, so it excludes higher-rank geometries by taking a rank-five residue. The conjugacy key is justified using actual conjugators and centralizers; no assertion that every automorphism of J_1 is inner is needed.

My fresh execution reproduced six constructions, 1536 marked intersections, 96 typed-Helly equalities, 100016 rank-four extensions and 21352 rank-five extensions. It found six rank-four classes and no rank-five classes. The complete class artifact agrees byte-for-byte with the expected certificate. The two negative geometric controls also reproduced their exact panel failures.

Generalized cubes. Danzer's construction, polytopality and original full automorphism group remain retained inputs. The shared graph-generation theorem identifies the even translation subgroup. The dual convention is handled explicitly: the exponent in the source counts facets of its input. The polygon counts, orientation and self-duality are proved, with Euler characteristic stated as a retained standard input.

The local-sphericity classification checks actual maximal residues, not merely their diagrams. The source lattice hypothesis excludes digonal sections. The finite Coxeter list leaves the stated simplex, cube and exceptional cases; the remaining F_4 possibility is excluded by an affine proper fork. Conversely the residues split into the proved D_j factor and the relevant spherical section. The exact toroidal identification is established by a Gray-map face-poset bijection, the distinguished affine reflections and the actual translation lattice $4\mathbb{Z}^r$. The rank-three square-map basis is checked separately. A group-order coincidence is not used as a substitute for these geometric identifications.

The convention from Ens, selected printed pages 70–75, is an inherited retained notation input. No stored primary PDF or text is available for that reading, so this review does not claim to have freshly checked those pages. The explicit lattice theorem in the application is nevertheless proved independently of an Ens classification. The comparison with every example in the source's reference [9] and its unused semiregular identification remain excluded. The rank-three limitation of the source's nonsphericity remark is stated.

Petrials. The new faces are actual pairs of vertex and edge sets, with equal pairs identified. The faithful-action theorem therefore gives the exact natural kernel, the core of the indicated order-two stabilizer. Residual connectivity is proved by replacing paths through old faces; it is not inferred from a diagram. The thinness obstruction treats separately whether an edge-changing involution preserves the Petrie face. The connected degree-two residue argument and the rotation/reflection calculation remain valid for infinite residues. After thinness has been proved, the action on new chambers and panel indices gives faithfulness. The second Petrial recovers old face shadows, and their distinctness is established explicitly. The C_2^3 and $C_2 \times S_3$ counterexamples correctly qualify the unconditional embedding and length assertions. No finite atlas or sampled family is used to prove the theorem.

Equal adjacent parameters. For every $n \geq 3$, the proposed model contains the pure dihedral factor and one edge translation, so shared graph generation gives the whole group. The complement-twist automorphism interchanges the endpoints, identifies both dihedral intersections, and supplies the marking reversal. The remaining maximal intersection is checked by translation coordinates. The proof then derives all subgroup intersections in rank three. The augmentation-basis adapter matches the source's actual matrices

and distinguished translation. The universal presentation uses the full range $1 \leq k < n$, and the common inverse construction proves injectivity. The $n = 2$ specialization uses its separate C_2^3 construction. The literal inverse-action multiplication in the source is correctly flagged as inconsistent with the stated left action. The earlier polygon construction receives attribution; no new existence or historical-priority claim is made.

5 Formal verification and remaining costs

Fresh compilation checked all three supplied modules: nine reported declarations in `SemidirectRecognition`, five in `BinaryDihedralDisplacements`, and two in `EndpointDoublingIntersections`. Every axiom report contains only the standard Lean axioms. The new module constructs the actual semidirect lift, proves surjectivity from subgroup generation, establishes both inverse identities and constructs the multiplicative equivalence. The reused modules prove their stated displacement and fixed-overlap identities.

The augmentation basis, the cyclic covariance instance, all source-specific incidence arguments, the GAP census and the source-to-proof correspondence remain written or computational obligations. They are not consequences of compilation alone. Missing pinned dependency objects were rebuilt privately; the remaining pinned package objects were trusted, not all recompiled. The retained catalogue and algorithm interfaces are likewise explicit mathematical dependencies.

The common paper is a full topic-local acquisition. Both complete mathematical GAP files are included once in the program appendix. They must not disappear from a later cost comparison. Operational drivers, formal validation and audit records are separately inventoried. The original L108 PDF is unavailable, so no four-source PDF-length numerator or complete-volume factor is authorized by this review. There is no claim about unmeasured graphic, historical program or retained external proof costs.